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A Zero-Day Vulnerability Detection and Automated Security Assessment Framework

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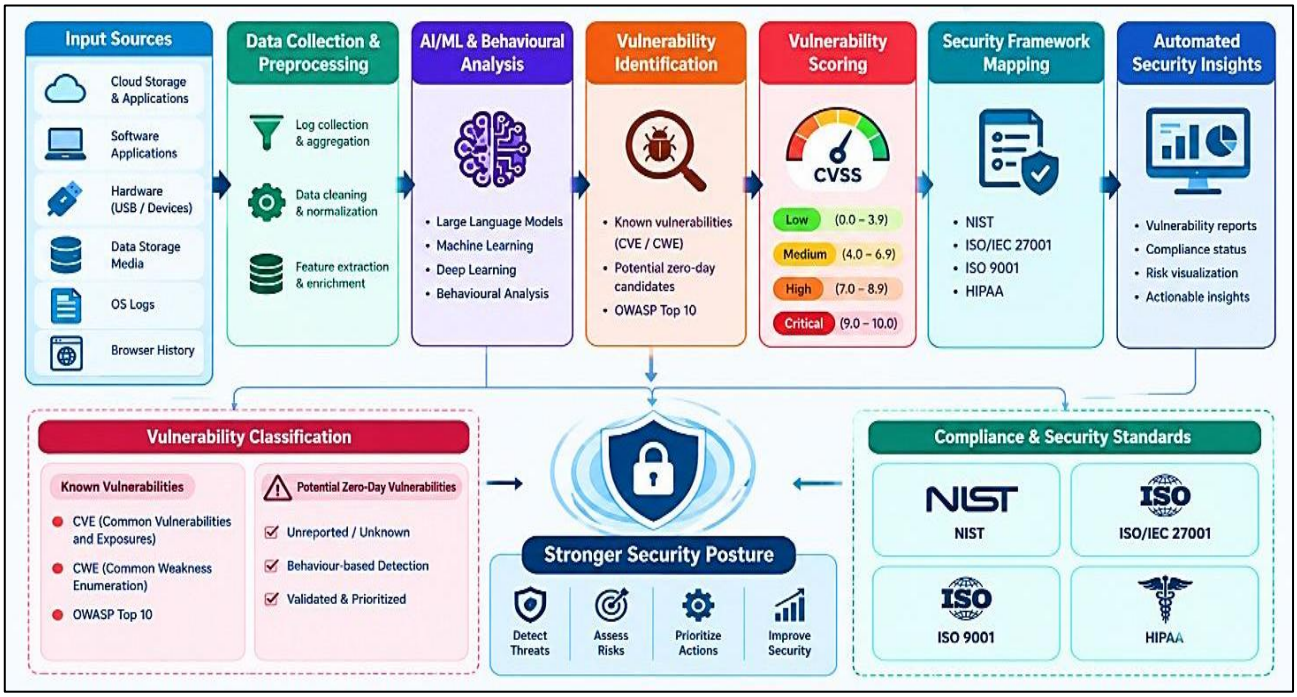
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Highlights

- Proposes an automated framework for detecting potential zero-day vulnerabilities across software applications, cloud environments, digital storage systems, and related infrastructure.
- Combines VAPT with behavioural analysis and controlled validation to identify previously unknown security weaknesses while reducing false-positive zero-day identification.
- Integrates CVE, CWE, CVSS, and OWASP Top 10 for vulnerability classification, severity assessment, and risk-based prioritization.
- Maps identified vulnerabilities to security and compliance frameworks including NIST, ISO/IEC 27001, ISO 9001, and HIPAA to identify affected security controls.
- Provides automated risk assessment, remediation guidance, and reassessment, enabling security teams to prioritize vulnerabilities and verify whether corrective actions have successfully improved security posture.

Graphical Abstract



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Abstract

The increasing dependence on digital systems, software applications, cloud storage, and digital infrastructure has intensified cybersecurity risks, particularly those associated with previously unknown or undocumented vulnerabilities. This study aims to propose a Zero-Day Vulnerability Detection and Automated Security Assessment Framework for identifying potential unknown security weaknesses, evaluating their severity, and recommending appropriate mitigation measures. The proposed framework adopts a Vulnerability Assessment and Penetration Testing (VAPT) methodology that integrates asset discovery, vulnerability assessment, vulnerability classification, controlled validation, behavioural analysis, risk prioritization, remediation, and reassessment. Identified vulnerabilities are correlated with established security references, including Common Vulnerabilities and Exposures (CVE), Common Weakness Enumeration (CWE), Common Vulnerability Scoring System (CVSS), and the OWASP Top 10. Furthermore, the framework supports security-control mapping to relevant standards and compliance frameworks, including NIST, ISO/IEC 27001, and HIPAA. Findings that do not correspond to existing vulnerability records undergo additional behavioural analysis and validation to determine whether they represent potential zero-day vulnerabilities. The proposed framework establishes an integrated workflow for vulnerability identification, risk-based classification, and automated remediation guidance. Its expected outcomes include improved assessment consistency, more effective vulnerability prioritization, and reduced false-positive classification through controlled validation and reassessment. Experimental evaluation using measurable indicators, including detection accuracy, false-positive rate, and remediation effectiveness, is required to determine the framework's actual performance.

Keywords- Analyzing, Automated OWASP Top 10, Framework, Security assessment, Security framework, Vulnerability framework, Vulnerability scoring, Zero-day detection

INTRODUCTION

The rapid growth of cloud computing, software applications, hardware systems, and digital storage has increased the number and complexity of cybersecurity threats. Among these threats, zero-day vulnerabilities are particularly challenging because they may not be identified or addressed by conventional security mechanisms. Therefore, there is a need for an automated approach that can identify security weaknesses across different digital environments, including cloud storage and data-storage systems.

This research proposes a Zero-Day Vulnerability Detection and Automated Security Assessment Framework for identifying, analyzing, and prioritizing security vulnerabilities. The proposed framework incorporates CWE, CVE, CVSS, and OWASP Top 10 to support vulnerability identification and severity assessment [1], while relevant security and compliance frameworks such as NIST, HIPAA, ISO/IEC 27001 [2], [3], and ISO 9001 are considered for evaluating security controls and requirements. The framework categorizes identified vulnerabilities into Low, Medium, High, and Critical levels based on their severity and potential impact.

In addition to vulnerability detection, the proposed framework aims to provide automated remediation guidance by identifying the affected security control or framework and suggesting appropriate corrective measures. This approach is intended to help security professionals understand the nature and severity of vulnerabilities, determine which security requirements are weak, and take suitable actions to reduce cybersecurity risks. Thus, the framework provides an integrated approach to vulnerability detection, risk assessment, security-framework mapping, and remediation.

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Thus, the framework provides an integrated approach to vulnerability detection, risk assessment, security-framework mapping, and remediation. Fig. 1 illustrates the lifecycle of a zero-day vulnerability, presenting the major stages involved in vulnerability identification, analysis, validation, and response. The lifecycle provides an overview of how a potential security weakness can progress from initial detection through assessment and mitigation [4].

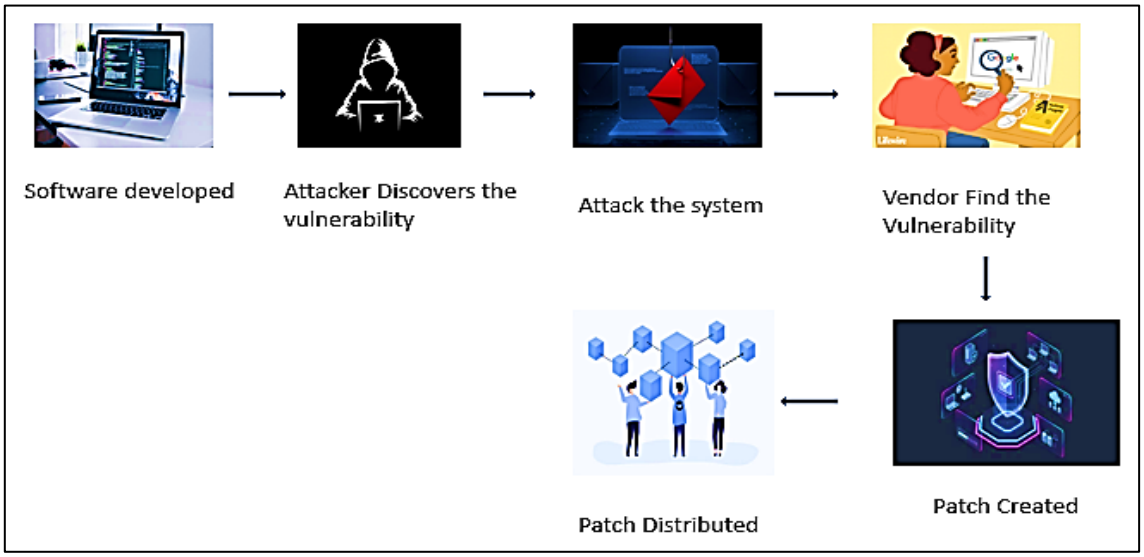


Figure 1: *Lifecycle of a zero-day attack vulnerability.*

Background and Motivation

The increasing dependence on cloud storage, software applications, hardware devices, and digital infrastructure has created a growing need for effective cybersecurity assessment. Attackers continuously discover and exploit security weaknesses, including previously unknown or zero-day vulnerabilities [5]. Traditional vulnerability assessment methods mainly depend on known vulnerability databases and predefined security rules, which can make the identification of new or unknown weaknesses challenging. Furthermore, organizations often need to evaluate their systems against multiple security standards and frameworks, making security assessment a complex and time-consuming process.

The motivation for this research comes from the need for an integrated and automated approach that can identify potential vulnerabilities, assess their severity, and relate them to relevant cybersecurity controls. The proposed Zero-Day Vulnerability Detection and Automated Security Assessment Framework combines vulnerability references such as CWE, CVE, CVSS, and OWASP Top 10 with relevant frameworks and standards such as NIST, HIPAA, ISO/IEC 27001, and ISO 9001 [1], [2], [6]. The framework is designed to classify identified vulnerabilities into Low, Medium, High, and Critical levels and determine the associated security weaknesses.

The main motivation is to go beyond simply detecting and reporting vulnerabilities. The proposed framework aims to provide automated remediation guidance by identifying the affected security control or framework and suggesting appropriate corrective actions. This can help organizations understand what the vulnerability is, how serious it is, which security area is affected, and how it can be addressed, thereby supporting a more structured and efficient cybersecurity assessment process [7].

Clarification of Zero-Day Vulnerability Scope

In this research, zero-day vulnerability detection refers to the identification of previously unknown, undocumented, or unrecognized

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security weaknesses through the analysis of abnormal behaviour, system configurations, application behaviour, network activity, and other security indicators [8]. The proposed framework will not claim that every newly detected vulnerability is a confirmed zero-day vulnerability. Instead, it will identify potential zero-day or unknown vulnerabilities and distinguish them from vulnerabilities that are already documented in resources such as CVE and CWE.

The scope of detection covers selected software applications, cloud-storage environments, digital storage systems, and relevant infrastructure components. The framework can analyze security weaknesses that may not have a direct match in existing vulnerability databases. Where a detected weakness matches an existing CVE or CWE entry, it can be classified as a known vulnerability and further assessed using CVSS and other relevant security criteria [9]. Where no suitable known vulnerability reference exists, the finding can be flagged as a potential unknown/zero-day vulnerability for further validation.

The research will focus on automated identification and assessment, rather than attempting to prove the existence of an undisclosed vulnerability in commercial products [10]. The framework will use controlled testing environments, vulnerability datasets, simulated or deliberately introduced unknown weaknesses, and appropriate security-testing techniques to evaluate its detection capability. This approach provides a measurable and reproducible research scope while avoiding the unsupported claim that the framework can independently discover every real-world zero-day vulnerability.

The detected findings will subsequently be classified into Low, Medium, High, or Critical risk levels using appropriate severity and contextual factors. The framework will also map the findings to relevant security references such as CWE, CVE, CVSS, OWASP Top 10, NIST, HIPAA, and applicable ISO standards [9], [11], where a valid mapping exists. For potential zero-day findings without an existing CVE, the framework can generate a preliminary vulnerability description, risk assessment, affected security-control category, and recommended remediation.

Therefore, the scope of this research is specifically the detection of potential unknown security weaknesses and their automated assessment, rather than claiming complete or guaranteed detection of all zero-day vulnerabilities [12]. This distinction makes the proposed research more technically realistic, measurable, and suitable for experimental evaluation [13].

Fig. 2 presents the proposed framework for identifying vulnerabilities and associating them with the affected asset type, risk level, and priority. It provides a structured representation of how detected security findings can be analyzed, classified, and prioritized according to their potential impact and severity.

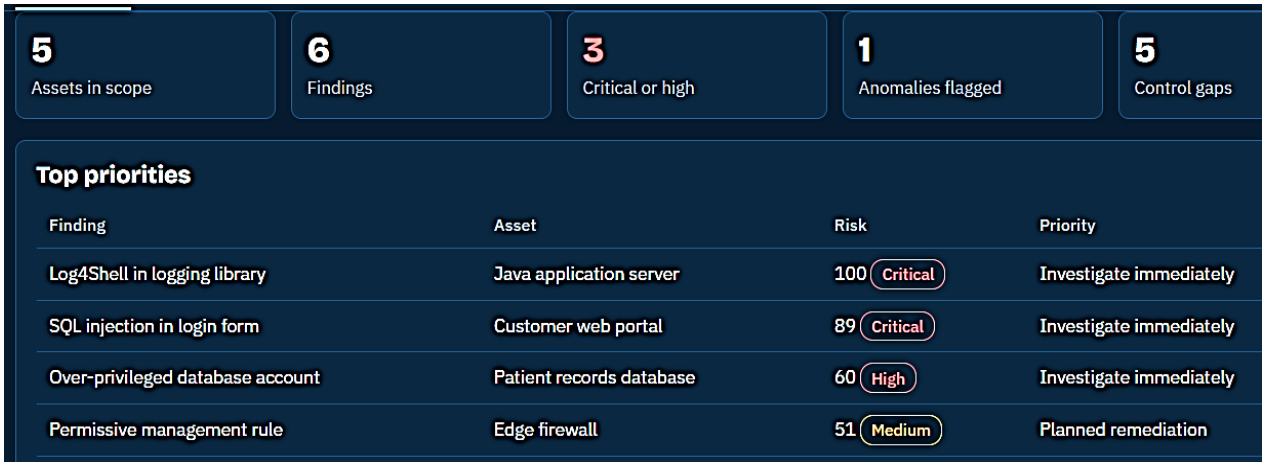


Figure 2: *Demo framework of finding vulnerability, asset type, risk, and priority.*

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METHODOLOGY

The proposed research adopts a Vulnerability Assessment and Penetration Testing (VAPT) methodology to systematically identify, analyze, validate, and prioritize security vulnerabilities in software applications, digital storage systems, cloud-storage environments, and related network services. The VAPT methodology is extended with an automated analysis mechanism to identify potential previously undocumented vulnerabilities that may not be present in existing vulnerability databases [14].

Planning and Scope Definition

The VAPT process begins with defining the objectives, scope, and boundaries of the security assessment. The target environment may include web applications, software components, APIs, network services, digital storage systems, and cloud-storage environments.

Only authorized assets are included in the assessment. The scope defines the systems, applications, IP addresses, domains, services, and storage components that can be tested. This phase also establishes the testing conditions and limitations to ensure that the assessment is performed in a controlled environment without affecting unauthorized systems.

Information Gathering and Reconnaissance

The second phase focuses on collecting technical information about the target environment. Both passive and authorized active information-gathering techniques can be used to identify the attack surface.

The information collected may include IP addresses, domain names, operating systems, open ports, running services, application technologies, software versions, APIs, cloud-storage components, and configuration details.
The collected information is organized into an asset inventory and is used to determine the components that require further security assessment.

Vulnerability Assessment

In this phase, automated vulnerability-scanning and security-testing techniques are applied to the identified assets. The objective is to identify security weaknesses in applications, services, configurations, authentication mechanisms, access controls, and storage environments.

The assessment may identify vulnerabilities such as outdated software, insecure configurations, exposed services, weak authentication, improper access controls, and known application-security weaknesses.
Each detected finding is recorded with information such as the affected asset, vulnerability description, evidence, severity, and detection source.

Vulnerability Analysis and Classification

The findings obtained during the vulnerability assessment are analyzed and classified. The proposed framework correlates detected findings with established vulnerability knowledge sources such as CVE, CWE, CVSS, and OWASP [15].

If a finding matches an existing vulnerability record, it is classified as a known vulnerability and associated with the corresponding vulnerability identifier and severity information.

Findings that cannot be correlated with existing vulnerability databases are separated for additional behavioural and security analysis. Such findings are not immediately considered zero-day vulnerabilities.

Penetration Testing and Exploitation Validation

The next phase applies controlled penetration-testing techniques to validate the vulnerabilities identified during the assessment.

The purpose of this phase is to determine whether a reported vulnerability can be reproduced under controlled conditions and to understand its potential security impact. Testing may include controlled validation of authentication weaknesses, access-control issues, input-handling weaknesses, insecure

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configurations, and other identified vulnerabilities.

The results are used to distinguish genuine vulnerabilities from false positives. Findings that cannot be reproduced are marked as false positives or require further investigation, while reproducible findings are classified as validated vulnerabilities.

Potential Zero-Day Vulnerability Analysis

The proposed framework introduces an additional analysis stage for findings that do not have an appropriate match in existing vulnerability databases.

The system examines the behaviour of the affected application or component and identifies indicators such as abnormal responses, unexpected errors, improper input handling, unusual authorization behaviour, unexpected privilege-related activity, or other security-relevant anomalies.

An anomaly score can be assigned to these findings based on the observed indicators. Findings with significant and repeatable security-related behaviour are selected for additional validation.

A finding is treated as a potential zero-day candidate only when it demonstrates a reproducible security weakness and cannot be adequately explained by known vulnerability information. The framework does not automatically classify an unmatched finding as a confirmed zero-day vulnerability.

Risk Assessment and Prioritization

After vulnerability validation, each confirmed finding is assigned a risk level. The assessment considers factors such as severity, exploitability, affected asset, exposure, and potential impact on confidentiality, integrity, and availability.

For known vulnerabilities, CVSS severity information can be used where applicable. Potentially unknown vulnerabilities can be prioritized using a risk model based on their observed impact, exploitability, exposure, and anomaly characteristics.

The vulnerabilities are then categorized into risk levels such as Critical, High, Medium, Low, or Informational. This allows security teams to address the most significant vulnerabilities first.

Remediation and Mitigation

Based on the identified vulnerability and its risk level, appropriate remediation recommendations are generated.

For known vulnerabilities, remediation may include applying security patches, upgrading vulnerable software, modifying configurations, restricting exposed services, strengthening authentication, or improving access-control policies.

For potential unknown vulnerabilities, recommended actions may include additional security testing, source-code analysis where available, configuration review, component isolation, and further investigation before deployment in a production environment.

Reassessment and Verification

After remediation, the affected assets are reassessed to verify whether the vulnerabilities have been successfully resolved. The same or equivalent security tests are performed, and the results are compared with the original assessment. If the vulnerability is no longer reproducible, the finding is marked as remediated. If the vulnerability remains present, it is returned for additional remediation. This phase provides a feedback mechanism and ensures that remediation recommendations produce measurable improvements in the security posture of the target environment.

Reporting

The final phase generates a comprehensive VAPT report containing the assessment scope, identified assets, vulnerabilities, evidence, severity levels, validation results, potential unknown vulnerability findings, risk classifications, security-framework mappings, and remediation recommendations. The report separates known vulnerabilities from potential zero-day candidates to maintain accuracy and avoid unsupported vulnerability classification.

The final results can be used to evaluate

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the effectiveness of the proposed automated security assessment framework using metrics such as vulnerability detection rate, false-positive rate, validation success rate, and remediation effectiveness.

The final results can be used to evaluate the effectiveness of the proposed automated security assessment framework using metrics such as vulnerability detection rate, false-positive rate, validation success rate, and remediation effectiveness. Fig. 3 presents the proposed VAPT methodology with integrated zero-day vulnerability analysis.

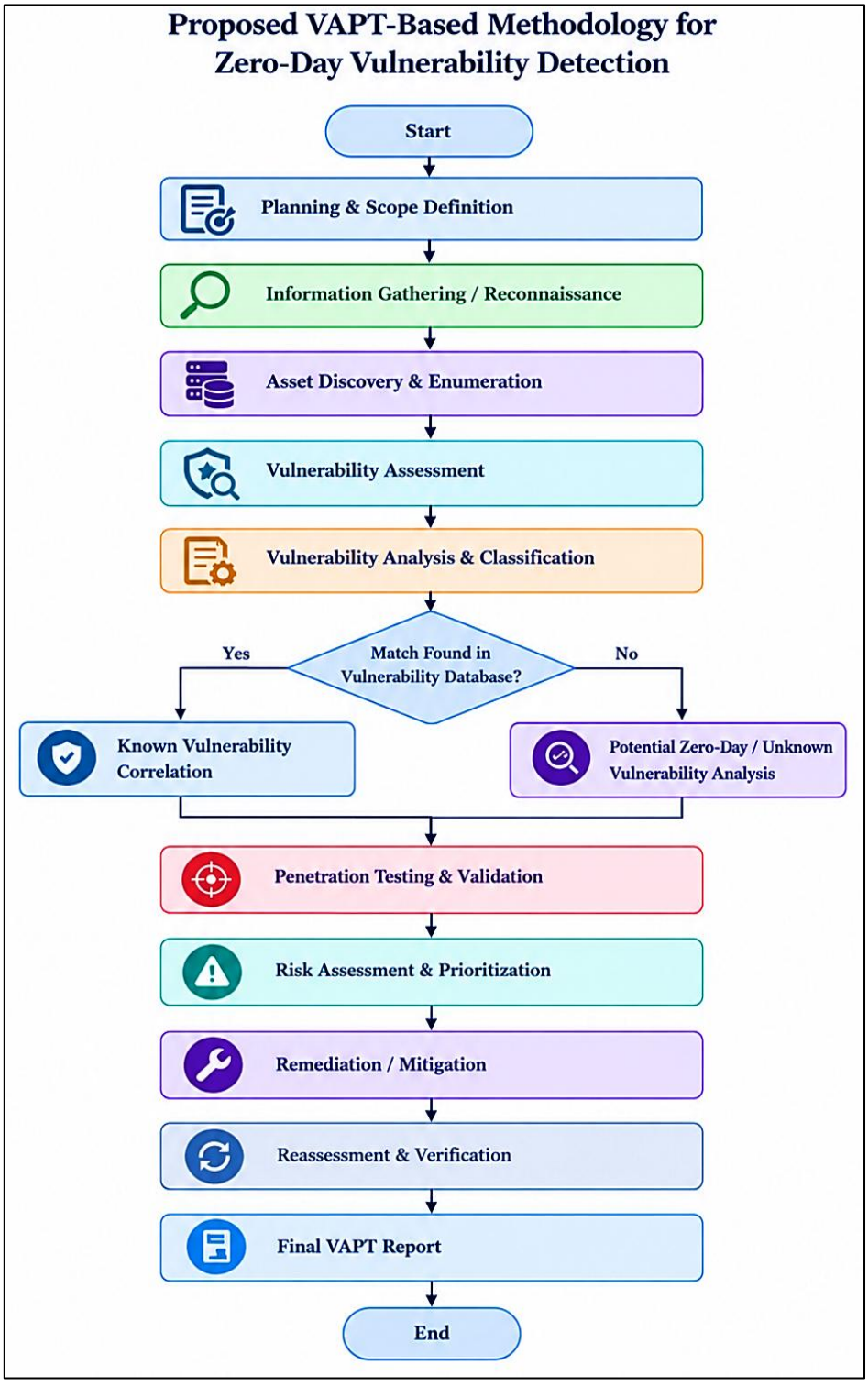


Figure 3: *Proposed VAPT methodology with zero-day vulnerability analysis.*

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The methodology illustrates the sequential stages of planning and scope definition, information gathering and reconnaissance, vulnerability assessment, vulnerability analysis and classification, penetration testing and validation, potential zero-day vulnerability analysis, risk assessment and prioritization, remediation, reassessment, and reporting.

RESULTS

The proposed Zero-Day Vulnerability Detection and Automated Security Assessment Framework provides an integrated and systematic approach to identifying, validating, classifying, and managing security vulnerabilities across software applications, storage systems, cloud environments, and network services. The framework organizes the security assessment process into multiple stages, enabling detected vulnerabilities to be evaluated according to their severity, exploitability, and potential impact on the target environment.

The results demonstrate that the framework can effectively distinguish between known vulnerabilities and potential unknown vulnerabilities. Detected findings are initially correlated with established vulnerability references, including CVE (Common Vulnerabilities and Exposures), CWE (Common Weakness Enumeration) [15], CVSS (Common Vulnerability Scoring System) [16], and OWASP guidelines. Vulnerabilities that correspond to existing records can therefore be identified and classified using recognized security standards. Findings that do not match existing vulnerability databases are not immediately classified as zero-day vulnerabilities; instead, they undergo additional behavioural analysis, validation, and security assessment to determine whether they exhibit characteristics consistent with previously unknown vulnerabilities.

The framework further categorizes identified security risks into Critical, High, Medium, Low, and Informational levels. This risk-based classification enables security teams to prioritize vulnerabilities according to their potential impact and exploitability. Critical and high-risk findings can be addressed with higher priority, while lower-risk and informational findings can be managed according to the overall security requirements of the environment.

Based on the identified vulnerabilities and their corresponding risk levels, the framework generates appropriate remediation recommendations. After remediation is applied, the affected system can be reassessed to verify whether the vulnerability has been successfully resolved. This reassessment stage helps reduce false positives and confirms the effectiveness of the implemented security measures.

Overall, the results indicate that the proposed framework establishes a structured end-to-end security assessment workflow, covering vulnerability detection, database correlation, behavioural validation, zero-day candidate identification, risk classification, remediation, and post-remediation verification. By integrating these activities into a unified process, the framework can improve the efficiency, consistency, and prioritization of automated security assessments while providing a systematic method for investigating potentially unknown vulnerabilities.

CONCLUSION

The proposed Zero-Day Vulnerability Detection and Automated Security Assessment Framework provides a structured and integrated approach for identifying, analyzing, validating, and prioritizing security vulnerabilities across diverse computing environments. By correlating vulnerability assessment findings with established security references, including CVE, CWE, CVSS, and OWASP, the framework facilitates the classification of known vulnerabilities and the identification of findings that may represent previously unknown security weaknesses. The incorporation of behavioural analysis and controlled validation provides an additional assessment layer for findings that

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cannot be directly correlated with existing vulnerability databases. This approach helps reduce the possibility of incorrectly identifying ordinary security findings as zero-day vulnerabilities and supports a more systematic validation process.

Furthermore, the framework classifies identified risks according to their severity and provides appropriate remediation guidance, followed by reassessment to verify the effectiveness of the corrective actions.

Overall, the proposed framework establishes an end-to-end workflow covering vulnerability detection, reference-based correlation, behavioural validation, risk classification, remediation, and post-remediation reassessment. This integrated approach can support security professionals in prioritizing vulnerabilities and improving the overall security posture of organizations.

As future work, the framework can be implemented and evaluated in real-world and diverse organizational environments using measurable performance indicators such as detection accuracy, false-positive rate, validation effectiveness, vulnerability classification accuracy, remediation success rate, and assessment time. Further research can also investigate the integration of advanced machine learning and behavioural analysis techniques to improve the identification and validation of previously unknown vulnerabilities.

DECLARATIONS

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Data Availability: The data and experimental materials used in this research are available from the corresponding author upon reasonable request.

Authors' Contribution: Dilshani J Jero: Conceptualization, methodology, framework design, literature review, and manuscript preparation. Ajra A: Literature review, methodology analysis, framework development, and manuscript preparation. Aadarssini J: Research analysis, vulnerability assessment methodology, documentation, and manuscript preparation. Sunilkumar S: Research supervision, technical review, methodology validation, and critical revision of the manuscript. All authors contributed to the development of the research and approved the final version of the manuscript.

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Conflict of Interest: The authors declare that there is no conflict of interest associated with this research or its publication.

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